Sampling and Intended Use

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Sampling and Intended Use

> Prepare Sampling and Intended Use: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Inspect row meaning **missing value**s and repeated records

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-4/5, ai-4/7, ai-4/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Write an observable label rule with boundaries and a review option

Reference example, not the only acceptable wording: Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible.

A partly hidden lane requires review because clear cannot be established from the available description.

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Specify a table **schema**

Reference example, not the only acceptable wording: activity: nonempty text; minutes: integer from 0 to 60 inclusive.

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish **fitting** a rule from applying a fixed rule

Reference example, not the only acceptable wording: The **input** is width x and the label is the observed category 0 or 1. Candidates are 2, 4 and 6.

Selection uses mistakes on the four **training** cards only; held out labels must not guide it.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

Concept to explain

Training examples should represent the people, objects, conditions, and devices described by the intended-use statement.

Term	Meaning
population	full group the product may encounter
sample	examples selected from that population
representation	coverage of relevant conditions
intended use	specific approved purpose and setting

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach inspection before modelling in two 45-minute sessions. Use fictional data only. Emphasize the unit of **observation**, **provenance**, units, unknown values and overlapping audit flags.

The **dataset card** is a concise classroom documentation exercise inspired by datasheets for datasets, not a claim to implement every part of a research framework.

Students must distinguish **observation**s from unsupported repairs and from **evidence** of model fitness.

Readiness answer

It is a numerical quantity whose differences measure time. Whole-number formatting does not turn it into a category.

Printed lesson and workbook

Cards for the five raw rows including both B rows

Optional Python 3 editor; no packages required

Blank card headings: purpose source row **schema** findings changes limits version owner

End goal for the pictorial case

By the end, I can explain sampling and **intended use** using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Sampling and Intended Use using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Compare two **sample**s against a deployment **population** and identify missing conditions.

MODEL IDEA **Training** examples should represent the people, objects, conditions, and devices described by the intended-use statement.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD **Intended use** includes indoor classroom plants under day and evening light. **Sample** A has 90% daylight photos from one camera.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 **Intended use** includes indoor classroom plants under day and evening light.

Step 2

STEP 2 **Sample** A has 90% daylight photos from one camera.

Step 3

STEP 3 **Sample** B balances lighting, camera distance, and several plant types.

Step 4

STEP 4 Prefer B for this use, while documenting remaining gaps rather than claiming universal coverage.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Training photos use indoor white backgrounds; **intended use** is outdoors.

The teaching case

Identify the coverage gap.

Exact explanation to model

> ### Why the result follows
>
> Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Explain the added worked example

Explain the added worked example · 1

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Explain the added worked example · 2

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

Explain the added worked example · 3

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

A successful run does not certify data quality.

Key terms

Term	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero
Dataset card	A concise record of dataset contents creation intended use and limitations

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Model the reasoning aloud · 5

Add **intended use**, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling and Intended Use

Sample for the conditions in which the plant classifier will actually be used.

The facts we will use

> #### Supplied facts
>
> **Training** set: 12 windowsill plants photographed at midday. **Intended use** includes evening desk photos; no evening examples have been collected.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/7

Sampling and Intended Use

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare coverage

All 12 training examples share one lighting situation.

2. Identify gap

Evening desk lighting is outside the observed sampling conditions.

3. Plan comparison

Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

CHANGE ONE THING

Collect 12 more midday windowsill photos of the same plants.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare coverage	All 12 training examples share one lighting situation.
2. Identify gap	Evening desk lighting is outside the observed sampling conditions.
3. Plan comparison	Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

Change one thing in the pictured case

Change one thing

Change: Collect 12 more midday windowsill photos of the same plants.

Predicted result

Prediction: The evening-coverage gap remains.

Why the result follows

Reason: More rows in the same condition do not supply **evidence** about the missing condition.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Sampling and Intended Use, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Training** photos use indoor white backgrounds; **intended use** is outdoors.

2. State the decision to explain

Write the question to explain: Identify the coverage gap.

3. Reveal the worked reasoning

Reveal after **prediction**s: Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Hold back the independent task

Keep for an independent attempt: **Training** photographs all show mature plants in bright light, but **intended use**rs photograph seedlings indoors. What claim is unsupported and what data is needed?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: 2 of 5 raw rows have unknown duration, or 40 percent. No: the extra B occurrence is already among the missing rows, so subtracting all three flag counts double-counts it.

A distinct usable-trip count requires a stated resolution policy and source review. Exit examples include duplicate IDs, impossible values, units, **provenance** and coverage.

A defensible statement is that this five-row fictional export has documented quality flags and is suitable for inspection practice, with no model-readiness claim.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve regression target meaning and why later outcomes are separate from prediction inputs. Establish that this week inspects data before fitting.

Stage	Teacher action
7 to 18 minutes	Read the source description and schema. Count physical rows and distinct trip IDs separately. Ask whether a repeated ID proves the reason for duplication.
18 to 32 minutes	Trace the code with row cards. Explain set, None and negative checks. Complete W1 and W2 with named denominators.
32 to 45 minutes	Use the guided question to show overlap: place both B cards in missing and repeated-ID groups. Do not subtract all flag counts independently.
Session 2 start to 8 minutes	Retrieve raw version versus corrected version. Ask what evidence could justify changing a recorded value.
8 to 20 minutes	Read the card example and complete W3. Distinguish a documented fact, an unknown and an assumption.
20 to 35 minutes	Collect W4 independently. Require counts, overlap and a concise card with source and limitations before offering hints.
35 to 45 minutes	Discuss W5 and the exit claim. Use the fresh retry after support and record the support given.

Reduce help deliberately

Completion practice

Completion task: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Reasoning to listen for

Teacher check: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Collect suitable evening desk **observation**s, with plant grouping and labels recorded, rather than only repeating midday views. Changed-case check: The evening-coverage gap remains.

More rows in the same condition do not supply **evidence** about the missing condition.

Changed-case reasoning: The evening-coverage gap remains. More rows in the same condition do not supply **evidence** about the missing condition.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Sampling and Intended Use investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Sampling Audit: compare condition percentages, map them to **intended use**, and propose the smallest justified collection plan.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A large dataset is representative simply because it contains many examples.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A complete card certifies data quality: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Missing means zero: None is unknown; zero is a specific allowed value in this exercise. Substitution without **evidence** changes the meaning.

Flag counts can always be added: The same row can have multiple flags. Count rows or unique events under an explicit policy and retain the raw denominator.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Sampling and Intended Use questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Independent question

Independent question · withhold the key

Training photographs all show mature plants in bright light, but **intended use**rs photograph seedlings indoors. What claim is unsupported and what data is needed?

Expected independent answer

Teacher key · Expected independent answer

Reliable performance on indoor seedlings is unsupported. Collect representative, appropriately permitted examples from the intended conditions and evaluate those groups.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Large data volume guarantees coverage of every **intended use**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Evaluate outdoor lighting conditions and different devices, with reliable labels and adequate counts. More same-setting photos may improve indoor fit but leave the deployment shift untested.

Record background and object variation without treating every difference as the proven cause.

Explanations for the additional workbook examples

W1 Define the row and source

Workbook question · W1

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

Teacher reasoning · W1

One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

W2 Interpret the audit counts

Workbook question · W2

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Teacher reasoning · W2

The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

W3 Reject an unsupported correction

Workbook question · W3

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

Teacher reasoning · W3

Those edits invent outcomes without source **evidence**. Preserve the raw version, flag the affected rows and investigate.

Record any supported correction with old value, new value, reason and version; unresolved values remain uncertain.

W4 Inspect and document independently

Workbook question · W4

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

Teacher reasoning · W4

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the duplicate and missing flags overlap. Zero for T is allowed by the stated **schema**.

The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no **evidence** for real or outdoor navigation use.

W5 State an evidence gap

Workbook question · W5

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

Teacher reasoning · W5

The row count increased but no new trip conditions or independent **observation**s were added. Coverage remains limited to the same supplied fictional indoor cases; the larger count does not establish broader usefulness.

Assess independent transfer

Independent question · withhold the key

Selected case: **Training** plant photos are all indoors on one phone. **Intended use** includes outdoor photos on other cameras. Plan a sampling check before making a general accuracy claim.

Selected case answer

Teacher key · Selected case answer

Collect and evaluate relevant outdoor and camera conditions, recording source and condition coverage. Group related views of each plant when splitting.

More indoor photos alone do not establish outdoor performance; report the limits of measured slices.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Sampling and Intended Use to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Freeze the Plant Classifier intended-use statement and a coverage table for species, lighting, camera, and health class.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 8: Leakage. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use physical row cards and overlapping circles for flags. Supply blank card headings but no completed findings.

Extension: explain how to record an **evidence**d seconds-to-minutes conversion while preserving the raw value, unit and version, and why a unit guess is not enough.

Require W4 counts 6,5,2,1,1 in the stated order, the Q overlap, acceptance of zero under this **schema**, and a card with source, unit, raw version and excluded use. Fresh retry: rows (L,3), (M,None), (N,-2), (N,-2), (O,0).

Counts are 5 raw, 4 IDs, 1 missing, 2 negative and 1 extra ID occurrence. The negative and repeated-ID flags overlap on N. Do not infer whether the two N rows describe one event without checking the source.

Week 5 examines label definitions and disagreement. Carry forward the **dataset card**, source questions and explicit distinction between unknown and observed values.

Teacher reference: https://arxiv.org/abs/1803.09010 (Datasheets for Datasets; reviewed 2026-09-26).

Plan the delayed return

Suggested return points in this level: ai-4/8, ai-4/10, ai-4/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Sampling and Intended Use lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Inspect row meaning **missing value**s and repeated records

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Supported practice answer

Reasoned practice answer: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Check every learner before discussion: Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Collect 12 more midday windowsill photos of the same plants. Require a **prediction** and a reason before comparing. Expected result: The evening-coverage gap remains.

Reason: More rows in the same condition do not supply **evidence** about the missing condition.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

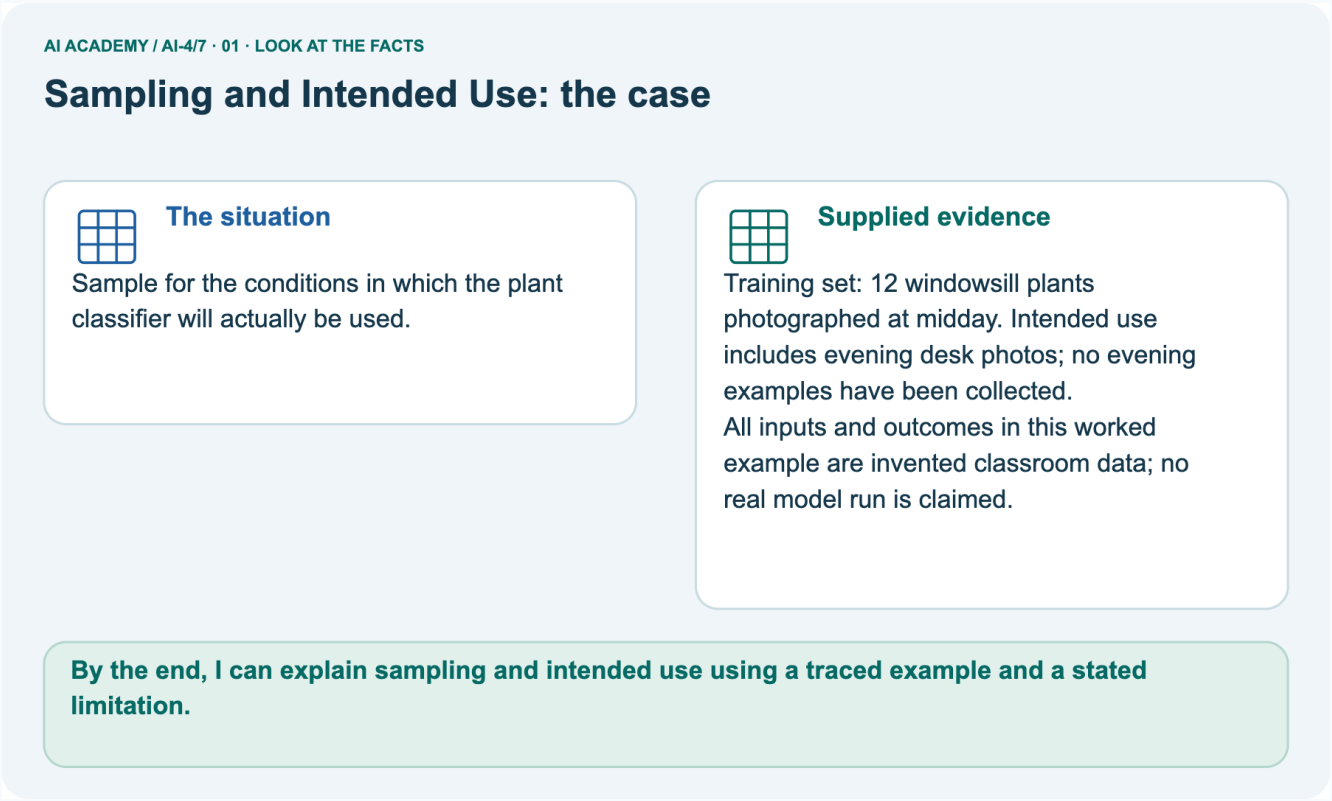
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Sampling and Intended Use graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sampling and Intended Use: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Sample for the conditions in which the plant classifier will actually be used. Supplied facts: **Training** set: 12 windowsill plants photographed at midday.

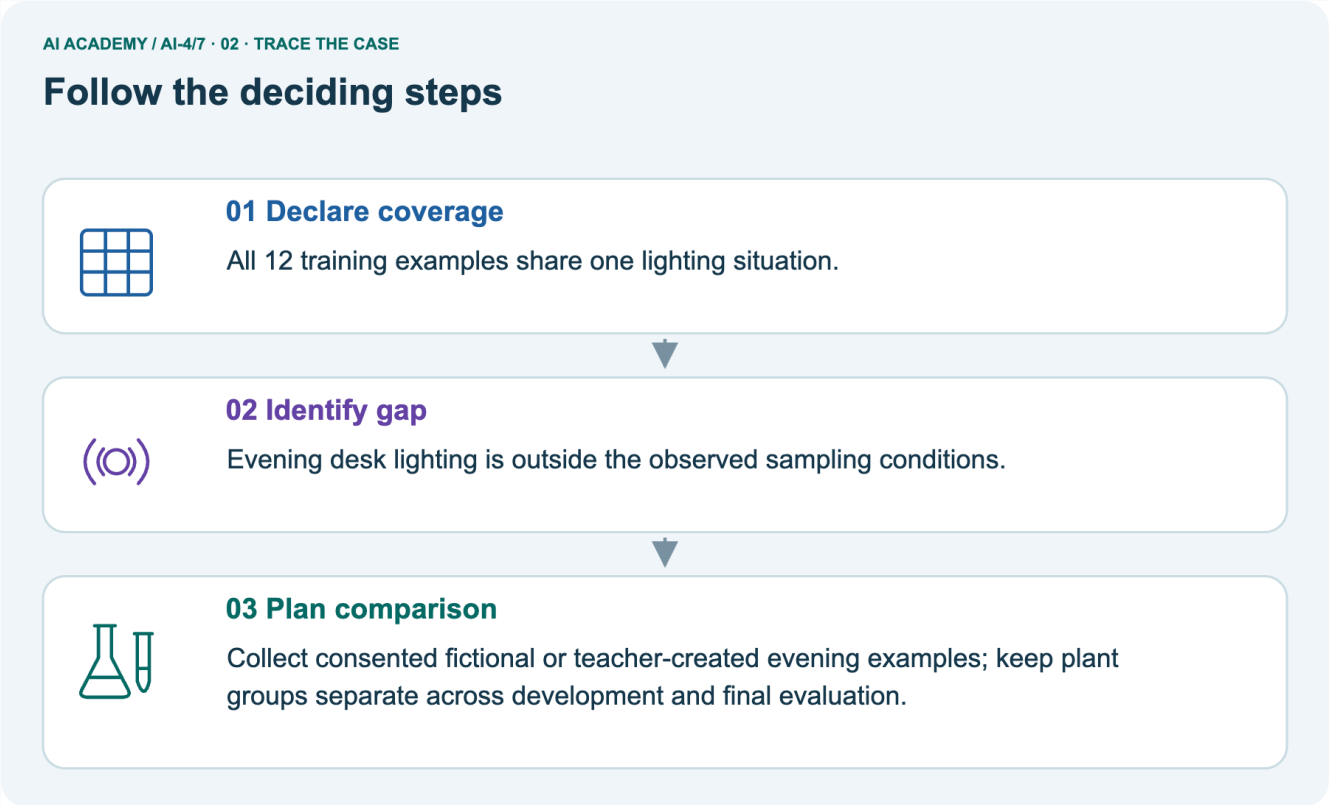
Intended use includes evening desk photos; no evening examples have been collected. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

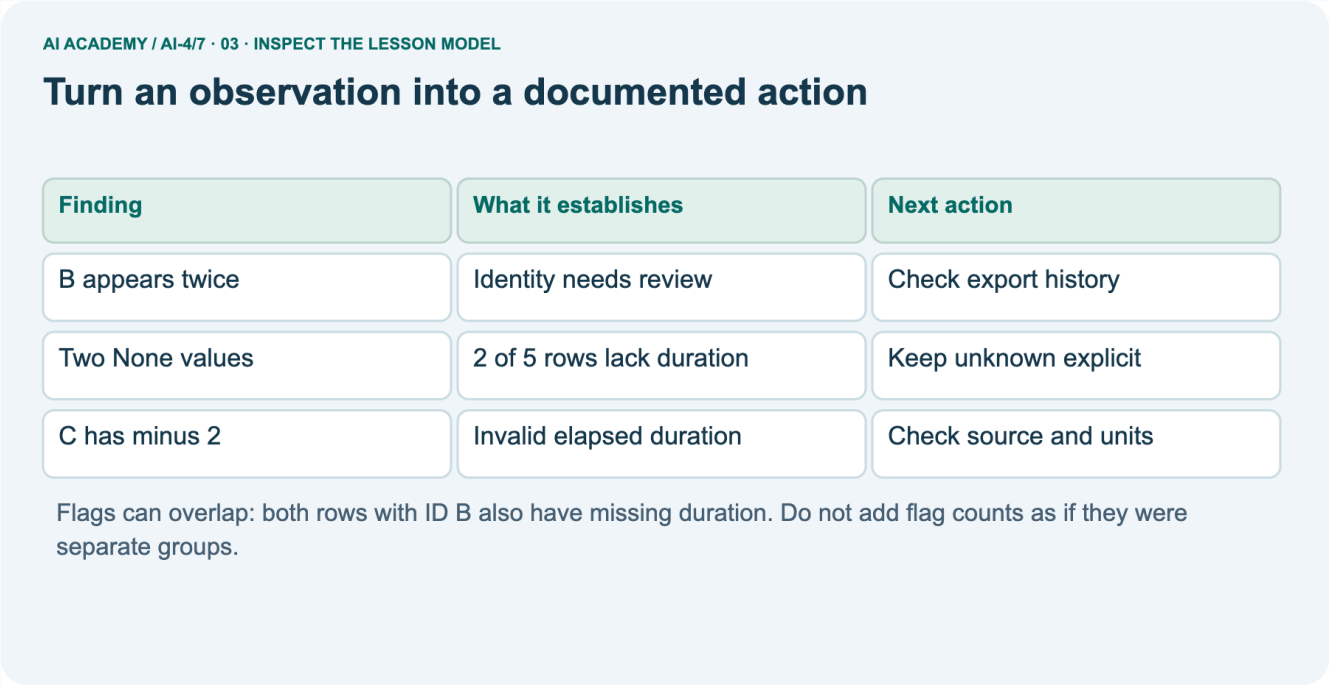
Declare coverage: All 12 **training** examples share one lighting situation. → Identify gap: Evening desk lighting is outside the observed sampling conditions. → Plan comparison: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

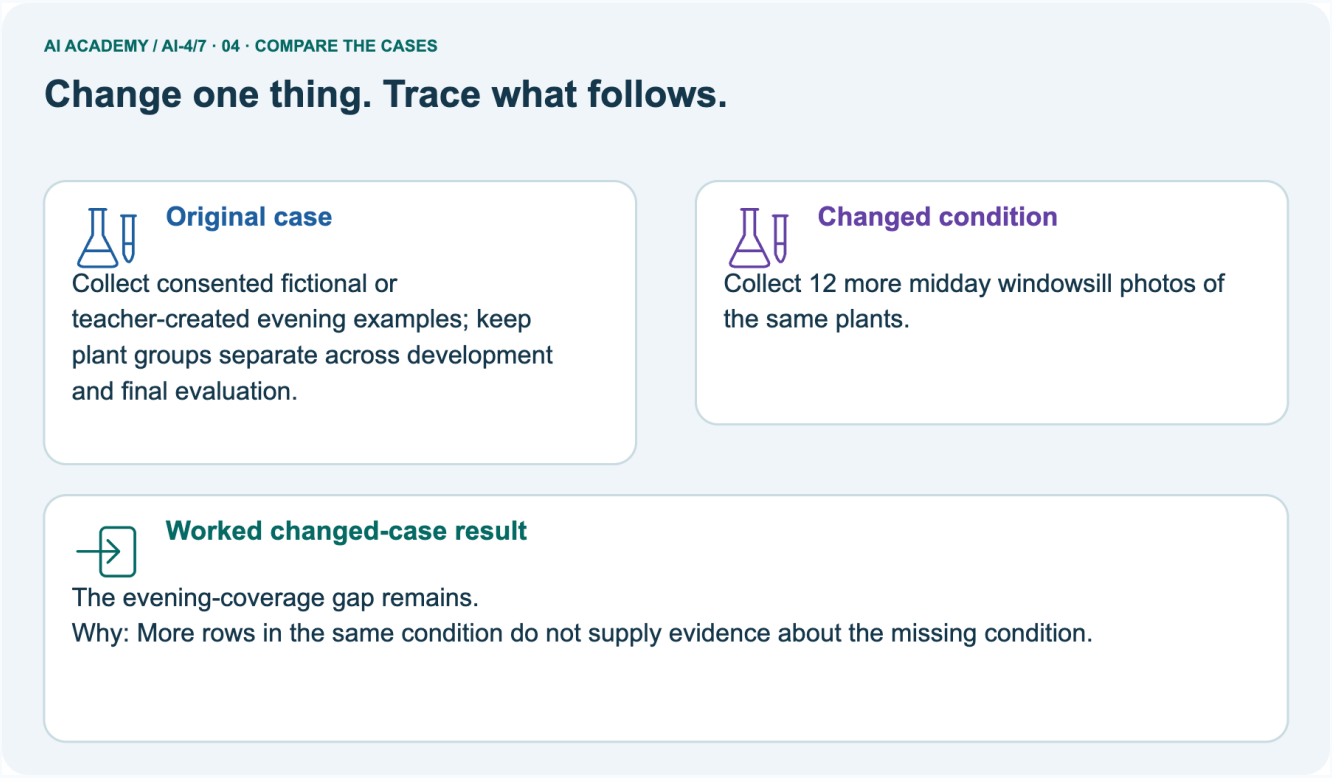
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

> Read the labels and values. Explain the relationship the model shows.

> **Discuss while showing the picture**
>
> Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Changed condition: Collect 12 more midday windowsill photos of the same plants. Expected result: The evening-coverage gap remains. Reason: More rows in the same condition do not supply **evidence** about the missing condition.

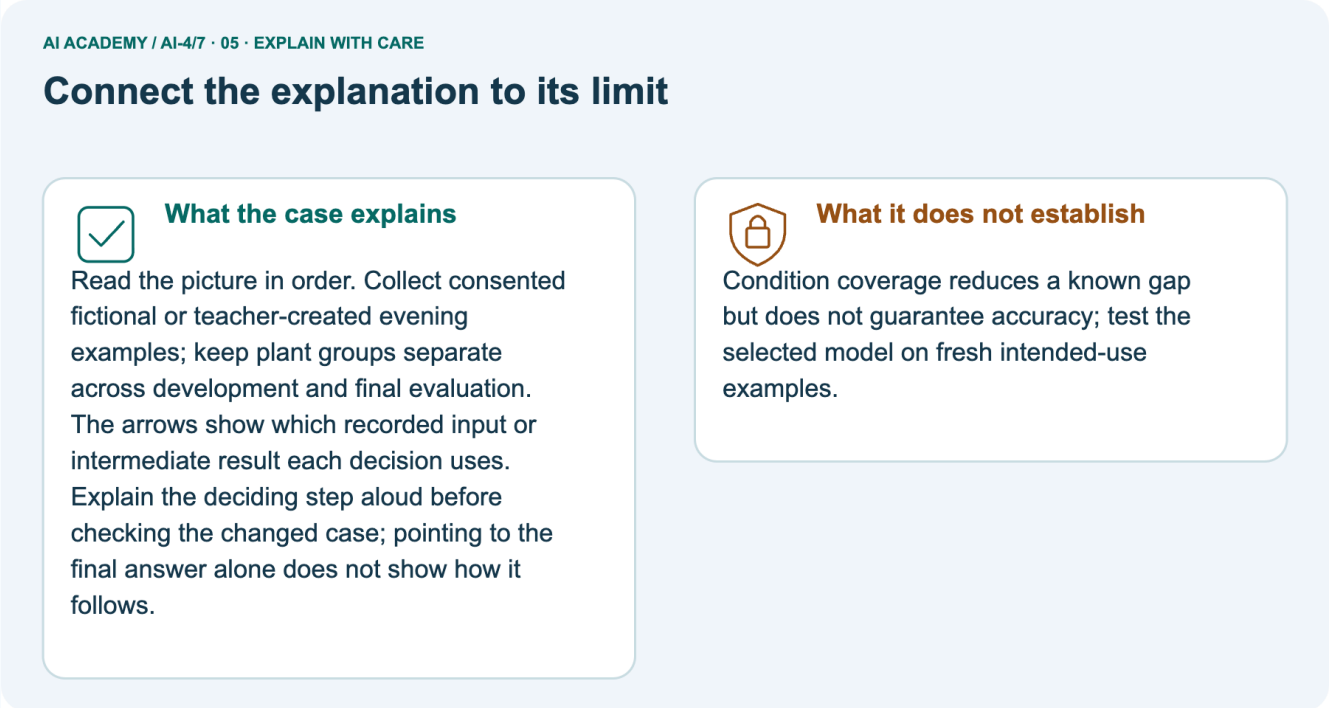
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: More rows in the same condition do not supply **evidence** about the missing condition.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

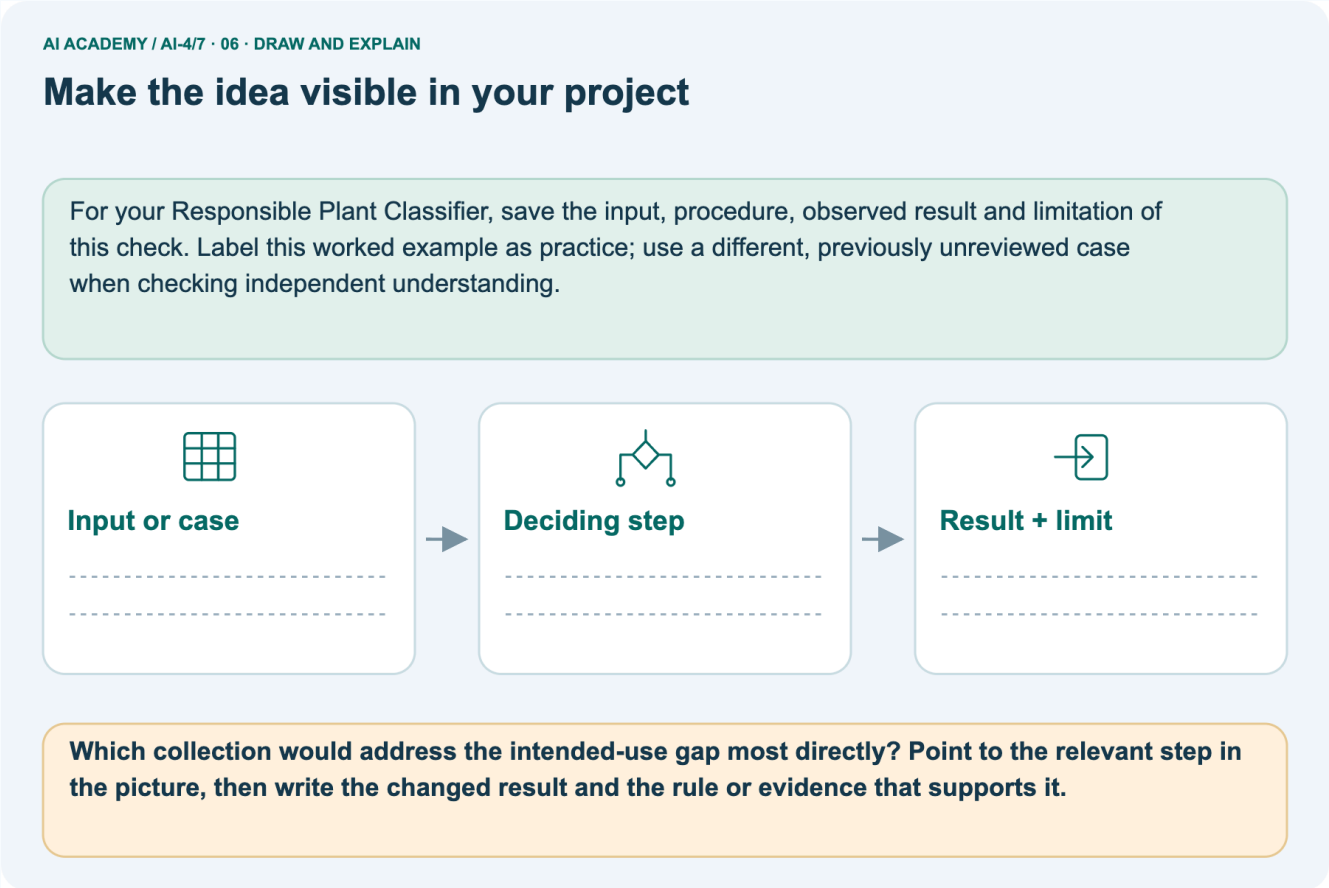
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Collect suitable evening desk **observation**s, with plant grouping and labels recorded, rather than only repeating midday views. Changed-case check: The evening-coverage gap remains.

More rows in the same condition do not supply **evidence** about the missing condition.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.